

PATTERN

```
import java.util.Scanner;

public class pattern {

    public static void main(String[] args){

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        for(int i=n; i>=n; i++){

            for(int j=1; j<=i; j++){

                System.out.print("* ");

            }

            System.out.println();

        }

        sc.close();

    }

}
```

ALPHABET PALINDROME PATTERN:

```
import java.util.Scanner;

public class alphabetspattern {

    public static void main(String[] args){

        Scanner sc = new Scanner(System.in);

        for(int i=3; i<=7; i++){

            for(int j=3; j<=i; j++){

                System.out.print((char)(j + 64) + " ");

            }

            System.out.println();

        }

        sc.close();

    }

}
```

ALPHABETPATTERN:

```
import java.util.Scanner;

public class alphabetpattern {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        for(int i=3; i<=7; i++){
            for(int j=3; j<=i; j++){
                System.out.print((char)(j + 64) + " ");
            }
            System.out.println();
        }
        sc.close();
    }
}
```

COUNT EVEN OR ODD:

```
public class countEvenOrOdd {
    public static void main(String[] args) {
        int[] arr = {12, 45, 7, 89, 23, 10, 8};
        int evenCount = 0;
        int oddCount = 0;
        for (int num : arr) {
            if (num % 2 == 0) {
                evenCount++;
            } else {
                oddCount++;
            }
        }
        System.out.println("Even numbers count: " + evenCount);
        System.out.println("Odd numbers count: " + oddCount);
    } }
}
```

COUNT FREQUENCY

```
public class countfrequency {  
    public static void main(String[] args) {  
        int[] arr = {10, 25, 90, 80, 25, 39, 42, 25, 81, 25};  
        int numberToCount = 25; // Change this to the number you want to count  
        int frequency = countFrequency(arr, numberToCount);  
        System.out.println("The frequency of " + numberToCount + " in the array is: " +  
frequency);  
    }  
    public static int countFrequency(int[] arr, int numberToCount) {  
        int count = 0;  
        for (int num : arr) {  
            if (num == numberToCount) {  
                count++;  
            }  
        }  
        return count;  
    }  
}
```

DIAMOND PYRAMID:

```
public class diamondpyramid {  
    public static void main(String[] args){  
        int n = 5;  
        for(int i=1; i<=n; i++){  
            for(int j=1; j<=n-i; j++){  
                System.out.print(" ");  
            }  
            for(int k=1; k<=i; k++){  
                System.out.print("* ");  
            }  
        }  
    }  
}
```

```

        System.out.println();
    }
    for(int i=n-1; i>=1; i--){
        for(int j=1; j<=n-i; j++){
            System.out.print(" ");
        }
        for(int k=1; k<=i; k++){
            System.out.print("* ");
        }
        System.out.println();
    }
}

```

EVEN PATTERN

```

import java.util.Scanner;

public class evenpattern {

    public static void main(String[] args){

        Scanner sc = new Scanner(System.in);

        int num = 0;

        for(int i=1; i<=4; i++){

            for(int j=1; j<=i; j++){

                System.out.print(num + " ");

                num+=2;

            }

            System.out.println();

        }

        sc.close();

    }

}

```

FULL PYRAMID:

```

import java.util.Scanner;

```

```

public class fullpyramidodd {

    public static void main(String[] args){

        Scanner sc = new Scanner(System.in);

        for(int i=1; i<=5; i++){

            for(int j=1; j<=5-i; j++){

                System.out.print(" ");

            }

            for(int k=1; k<=2*i-1; k++){

                System.out.print("* ");

            }

            System.out.println();

        }

        sc.close();

    }

}

```

MAXIMUM ELEMENT:

```

public class MaximumElement {

    public static void main(String[] args) {

        int[] arr = {12, 45, 7, 89, 23};

        int max = arr[0];

        for (int i = 1; i < arr.length; i++) {

            if (arr[i] > max) {

                max = arr[i];

            }

        }

        System.out.println("Maximum element = " + max);

    }

}

```

MINIMUM ELEMENT:

```
import java.util.Scanner;
```

```
public class min {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter the size of the array: ");
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
        System.out.println("Enter the elements of the array:");
```

```
        for (int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        int min= findMin(arr);
```

```
        System.out.println("The minimum element in the array is: " + min);
```

```
    }
```

```
    public static int findMin(int[] arr) {
```

```
        int min = arr[0];
```

```
        for (int i = 1; i < arr.length; i++) {
```

```
            if (arr[i] < min) {
```

```
                min = arr[i];
```

```
            }
```

```
        }
```

```
        return min;
```

```
    }
```

PATTERN NUMBERS:

```
import java.util.Scanner;
```

```
public class patternnos {
```

```
public static void main(String[] args){  
    Scanner sc = new Scanner(System.in);  
    for(int i=3; i<=7; i++){  
        for(int j=3; j<=i; j++){  
            System.out.print(j + " ");  
        }  
        System.out.println();  
    }  
    sc.close();  
}
```

REVERSE ARRAY:

```
public class ReverseArray {  
    public static void main(String[] args) {  
        int[] arr = {23, 29, 45, 49, 33, 28, 11};  
  
        System.out.println("Original array:");  
        printArray(arr);  
  
        reverseArray(arr);  
  
        System.out.println("Reversed array:");  
        printArray(arr);  
    }  
  
    public static void reverseArray(int[] arr) {  
        int left = 0;  
        int right = arr.length - 1;
```

```

while (left < right) {

    // Swap elements at left and right indices

    int temp = arr[left];
    arr[left] = arr[right];
    arr[right] = temp;

    // Move towards the middle

    left++;
    right--;
}
}

```

```

public static void printArray(int[] arr) {
    for (int num : arr) {
        System.out.print(num + " ");
    }
    System.out.println();
}
}

```

SECOND LARGEST NUMBER

```

public class SecondLargestNumber {
    public static void main(String[] args) {
        int[] arr = {12, 45, 7, 89, 23, 10, 8};

        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;

        for (int num : arr) {
            if (num > largest) {
                secondLargest = largest; // Update second largest before updating largest
                largest = num; // Update largest
            }
        }
    }
}

```



```

    } else if (num > secondLargest && num != largest) {
        secondLargest = num; // Update second largest if it's greater than current second
largest and not equal to largest
    }
}

if (secondLargest == Integer.MIN_VALUE) {
    System.out.println("There is no second largest number in the array.");
} else {
    System.out.println("The second largest number in the array is: " + secondLargest);
}
}
}

```

ZEROS AND ONES:

```

import java.util.Scanner;

public class zeroandonespattern {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        for(int i=1; i<=4; i++){
            for(int j=1; j<=i; j++){
                System.out.print((i + j) % 2 + " ");
            }
            System.out.println();
        }
        sc.close();
    }
}

```

